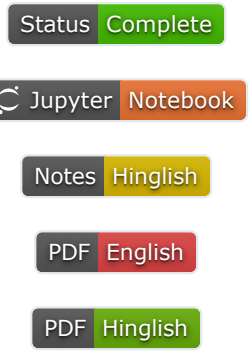
Lecture 31.12: Hierarchical Clustering (Code) — Scipy & Scikit-Learn



📌 Topic Overview

Practical implementation of Hierarchical Clustering using scipy and scikit-learn. Demonstrates computing the linkage matrix with scipy.cluster.hierarchy.linkage, plotting dendrograms, and fitting sklearn.cluster.AgglomerativeClustering.

📋 Key Learning Objectives

- Importing linkage and dendrogram from scipy.cluster.hierarchy.
- Computing Ward's linkage matrix on standardized feature data.
- Plotting and styling the dendrogram with matplotlib figure size.
- Importing AgglomerativeClustering from sklearn.cluster.
- Fitting with n_clusters=3, metric='euclidean', linkage='ward' and visualizing partitions.

🔍 In-Depth Conceptual Breakdown

- **Scipy Linkage Matrix**: `from scipy.cluster.hierarchy import linkage, dendrogram` use kiya gaya. `linkage(X_scaled, method='ward')` function saare pairwise clusters ke merging sequence aur distance ki table banata hai.
- **Dendrogram Plotting**: `plt.figure(figsize=(12, 6))` banakar `dendrogram(linkage_matrix)` call kiya gaya. Plot mein clearly 2-3 prominent vertical branches nazar aati hain.
- **AgglomerativeClustering in Scikit-Learn**: `from sklearn.cluster import AgglomerativeClustering` use kiya gaya.

- **Model Setup**: `agg = AgglomerativeClustering(n_clusters=3, metric='euclidean', linkage='ward')` instantiate kiya gaya.
- **Fit & Predict**: `labels = agg.fit_predict(X_scaled)` se har point ko cluster label mila.
- **Visualization**: `sns.scatterplot(x=X_scaled[:, 0], y=X_scaled[:, 2], c=labels)` se 3 distinct clusters beautifully visualize huye.
- **Code Lecture**: Is lecture ka complete runnable notebook lecture_31_12.ipynb mein available hai.

📋 Scipy vs Scikit-Learn Hierarchical Functions

Task / Objective	Library Tool	Primary Output
Compute Merge Tree	`scipy.cluster.hierarchy.linkage`	$(N-1) \times 4$ Linkage Matrix
Visualize Tree	`scipy.cluster.hierarchy.dendrogram`	Dendrogram Matplotlib Tree Plot
Cluster Partitioning	`sklearn.cluster.AgglomerativeClustering`	Cluster Label Array $\hat{y} \in \{0, \ldots, K-1\}$

📊 Scipy + Sklearn Hierarchical Code Pipeline



🔢 Mathematical Formulation

Agglomerative cluster label assignment:

$$c_i = \text{AgglomerativeClustering}(K=3, \text{ward}).\text{fit_predict}(\mathbf{X})_i$$

💻 Implementation

The standalone verified code is provided in the synchronized Jupyter Notebook.